

Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Localhost Instance Thread Blockage	Single-threaded Flask development server stalls under multiple simultaneous application evaluations.	High
2	File-Based Binary Memory Allocation	Loading fraud_model.pkl directly into runtime memory consumes excessive RAM during scale out requests.	Medium
3	Stateless Data Payload Pipeline	Incoming client applications are processed in-memory but not persistently logged to a permanent database ledger.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single evaluation streams locally.	Deploy application container boundaries behind a cloud-managed load balancer (AWS/GCP infrastructure).
2	Data Storage	No persistent data repository integrated.	Link an enterprise relational database (PostgreSQL) to store transaction application metadata cleanly.
3	Performance	Synchronous file parsing calculations latency.	Implement a production WSGI application engine wrapper (Gunicorn) with multiple asynchronous worker instances.
4	Security	Plain textual vector payload requests channel.	Encrypt communication pipelines via TLS/HTTPS protocols and implement JWT authentication token boundaries.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Production Server Wrapper & Containerization	Month 1	Eliminates local script crashing loops under persistent access traffic.
Phase 3	Relational Database Connection	Month 2	Logs input parameters for permanent historical analysis auditing.
Phase 4	Continuous Model Retraining Setup	Month 3	Keeps classification boundary layers updated against changing default trends.